FOODIQ

Decision Tree Model Evaluation Report

Milk composition screening model based on the project test dataset

Evaluation scope	Value
Model	DecisionTreeClassifier
Dataset	1,200 rule-derived milk-composition test records
Features	Fat, SNF, total sodium, total urea and milk class
Target	PASS or HOLD composition-screen result
Evaluation	80:20 stratified held-out test and 5-fold stratified cross-validation

Executive summary

The model achieved **95.42%** accuracy on the held-out test set and **90.08%** mean accuracy in 5-fold cross-validation. It correctly identified 64 of 72 HOLD records in the held-out set, giving a HOLD recall of **88.89%**.

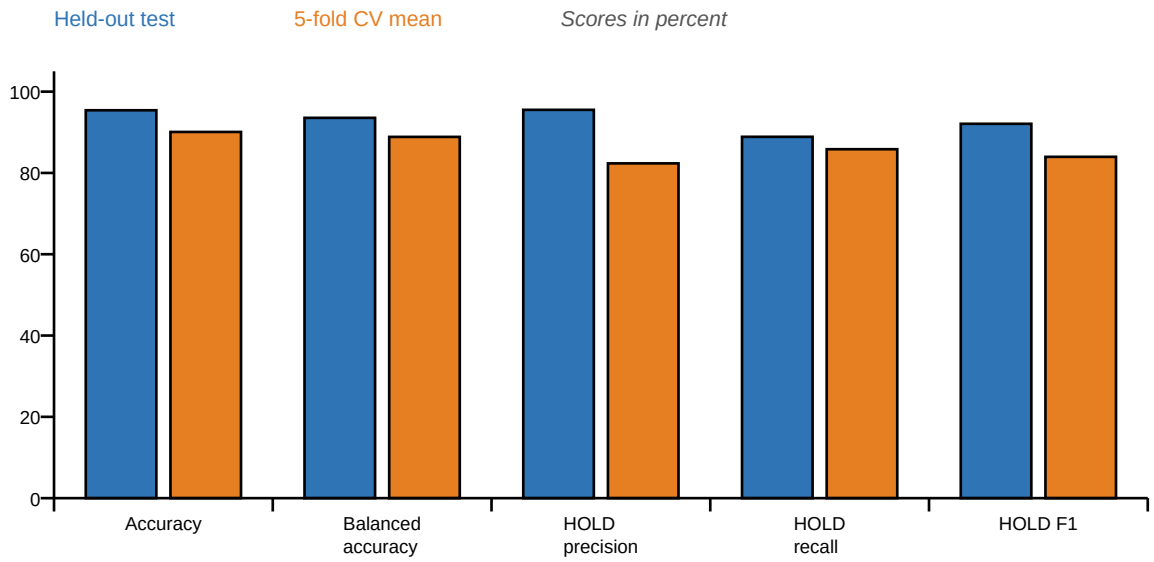
Interpretation: these results measure agreement with rule-derived test labels. The model is an in-app decision-support demonstration and does not replace FSSAI verification, accredited laboratory testing, or expert review.

Key held-out test results

Metric	Result	What it indicates
Accuracy	95.42%	Overall correct predictions across PASS and HOLD.
Balanced accuracy	93.55%	Average class-aware performance despite class imbalance.
HOLD precision	95.52%	How often a predicted HOLD was actually HOLD.
HOLD recall	88.89%	How many actual HOLD records were detected.
HOLD F1 score	92.09%	Balance between HOLD precision and recall.
PASS recall	98.21%	How many actual PASS records were detected.

Performance comparison

The chart compares the held-out test scores with the mean of five stratified cross-validation folds. Error bars on the orange bars show the cross-validation standard deviation.



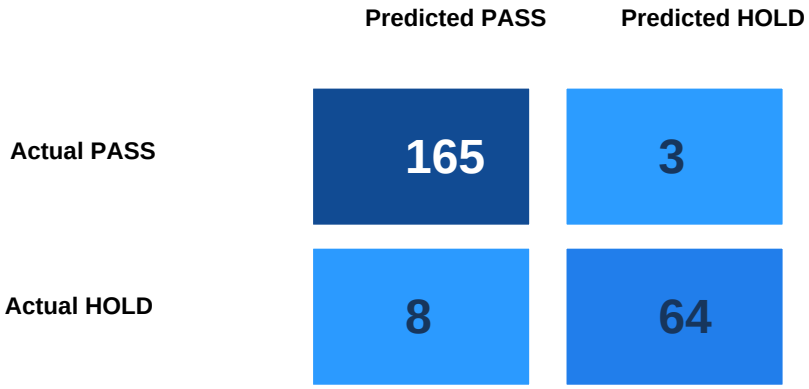
Cross-validation summary

Metric	5-fold mean	Standard deviation
Accuracy	90.08%	2.51%
Balanced accuracy	88.87%	2.07%
HOLD precision	82.35%	5.73%
HOLD recall	85.83%	1.84%
HOLD F1 score	83.96%	3.47%

Confusion matrix

Rows show actual labels and columns show predicted labels. Correct predictions lie on the diagonal.

Held-out Test Confusion Matrix



Rows = actual labels; columns = predicted labels

Model configuration and evaluation notes

Parameter	Configured value
Criterion	gini
Maximum depth	8
Minimum samples to split	20
Minimum samples per leaf	10
Class weighting	PASS: 1, HOLD: 2
Random state	42
Training / test split	960 / 240 records, stratified

Held-out confusion matrix values

Actual label	Predicted PASS	Predicted HOLD	Total
PASS	165	3	168
HOLD	8	64	72
Total	173	67	240

Reading the results

- The stronger PASS recall means that 165 of 168 PASS cases in the held-out split were recognised as PASS.

- Eight actual HOLD cases were predicted as PASS. In a quality-screening setting, this type of error is important and should be reviewed rather than treated as an automated approval.

- The model features intentionally exclude the rule-limit columns and failure-reason text, avoiding direct leakage of the generated label into the prediction.

- The exported project artifacts include the full metric JSON, a CSV confusion matrix, a readable tree-rules text file and the saved joblib model.

Source files used

reports/fssai_decision_tree_demo_metrics.json; reports/fssai_decision_tree_demo_confusion_matrix.csv; train_fssai_decision_tree_demo.py; models/fssai_decision_tree_demo.joblib.